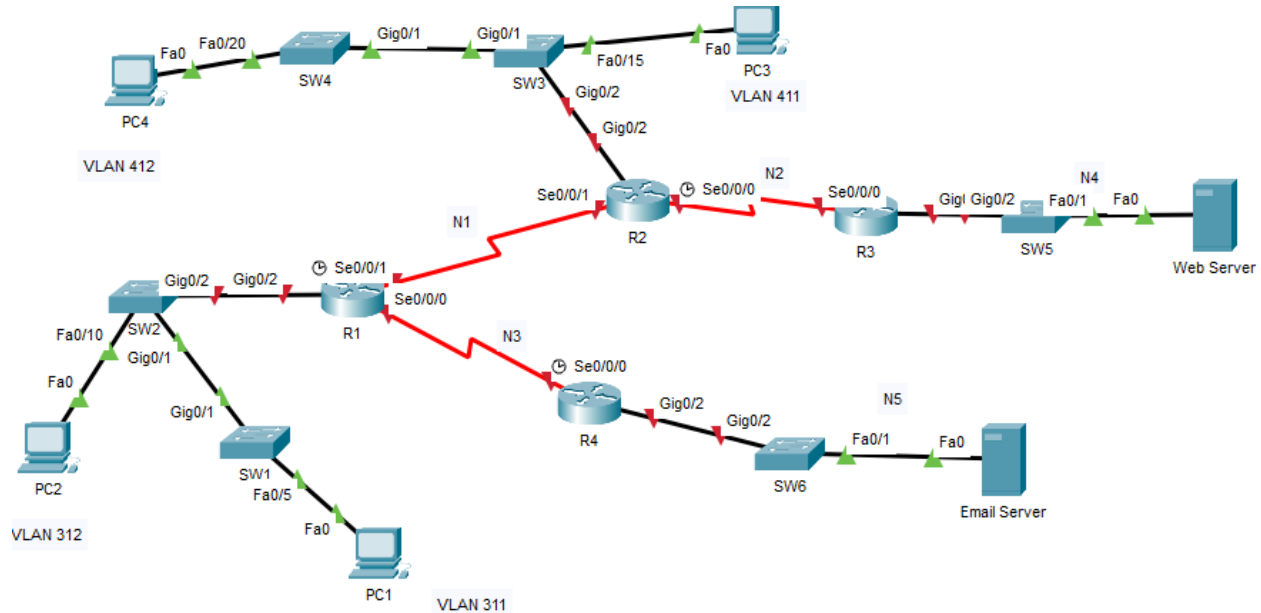


**WIA1005 Network Technology Foundation
Practical Test (15%)
(2 Hours)
Topology Diagram**



Network Address Table

Network	IPv4	IPv6
N1	183.21.2.244/30	2001:3:3:21::/64
N2	183.21.2.248/30	2001:3:3:22::/64
N3	183.21.2.252/30	2001:3:3:23::/64
N4	183.21.2.192/28	2001:3:3:24::/64
N5	183.21.2.160/27	2001:3:3:25::/64
VLAN 311	183.21.0.0/26	2001:6:6:311::/64
VLAN 312	183.21.0.64/26	2001:6:6:312::/64
VLAN 388	183.21.0.128/28	-
VLAN 411	183.21.1.0/25	2001:6:6:411::/64
VLAN 412	183.21.1.128/27	2001:6:6:412::/64
VLAN 488	183.21.1.160/28	-

VLAN Table (SW1, SW2, SW3, SW4)

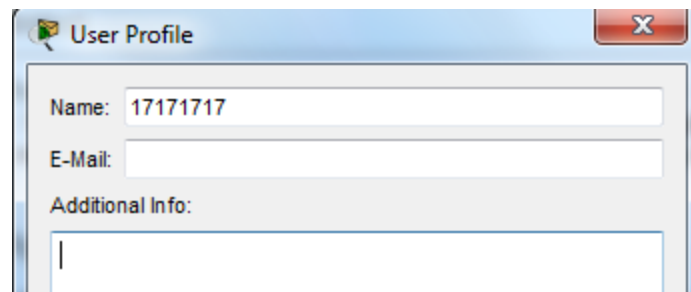
VLAN	Name	Device	Port
311	MOE-L1	PC1	SW1 Fa0/5
312	MOE-L2	PC2	SW2 Fa0/10
388	Management	SW1, SW2	-
411	MOSTI-L1	PC3	SW3 Fa0/15
412	MOSTI-L2	PC4	SW4 Fa0/20
488	Management	SW3, SW4	-

IP Address Table

Device	Port	IPv4	IPv6 Interface ID
R1	S0/0/0	First Host of N3	::2020
	S0/0/1	First Host of N1	::2020
	G0/2.311	First Host IP of VLAN 311	::2020
	G0/2.312	First Host IP of VLAN 312	::2020
	G0/2.388	Last Host IP of VLAN 388	-
R2	G0/2.411	First Host IP of VLAN 411	::2020
	G0/2.412	First Host IP of VLAN 412	::2020
	G0/2.488	Last Host IP of VLAN 488	-
	S0/0/0	Last Host of N2	::2021
	S0/0/1	Last Host of N1	::2021
R3	S0/0/0	First Host of N2	::2020
	G0/2	Last Host of N4	::2021
R4	S0/0/0	Last Host of N3	::2021
	G0/2	Last Host of N5	::2021
SW1	Management VLAN	First Host IP of VLAN 388	-
SW2	Management VLAN	Second Host IP of VLAN 388	-
SW3	Management VLAN	First Host IP of VLAN 488	-
SW4	Management VLAN	Second Host IP of VLAN 488	
SW5	SVI	Second Last Host of N4	-
SW6	SVI	Second Last Host of N5	-
Email Server	NIC	First Host IP of N5	::2020
Web Server	NIC	First Host IP of N4	::2020
PC1	NIC	DHCPv4	::2021
PC2	NIC	DHCPv4	::2021
PC3	NIC	Third Host IP of VLAN 411	::2021
PC4	NIC	Third Host IP of VLAN 412	::2021

Packet Tracer

Download the Packet Tracer **WIA1005-PT.pka** file from your email and then rename the file as **matricnumber.pka**, example: **17171717.pka** or **25001717.pka** or **S2101717.pka**. Enter your **matric number** in the name input when the user profile dialog box is shown. Please check whether you enter the correct matric number **before** you start the configuration. Failure to enter matric number will result in **marks deduction**.



Task 1:

Configure host name on all the routers and switches. Configure VLAN on SW1, SW2, SW3 and SW4; VLAN 388 and VLAN 488 are native VLANs. Then, configure the IP address on all the devices (PC, servers, switches and routers) based on the tables above.

Task 2: Switch Configuration

Configure latest SSH version on SW2. Given the domain name is **um.edu.my**. The length of RSA keys is **1024** and the username is **pensyarah** with the secret password **cisco**.

Secure the access port on SW1, SW2, SW3 and SW4 from MAC address flooding. Limit only **THREE** devices to access the port. The MAC address is dynamically learned and added to the running configuration. Disable the port when a violation occurs.

Task 3: DHCPv4 Configuration

Configure R1 as the DHCP server for VLAN311 and VLAN312. Create the DHCP pool named **VLAN311** and **VLAN312**. Reserve the **FIRST** IPv4 address and the last **THREE** IPv4 addresses of each VLAN for special purpose. Given the DNS Server is 183.21.0.136.

Task 4: Static Route Configuration (IPv4, IPv6)

- Configure IPv4 and IPv6 static routes in R1 and R2 using the next hop command. You must summarize the IPv4 VLAN311, VLAN312 and VLAN388 using one network 183.21.0.0/24 while VLAN411, VLAN412 and VLAN488 using one network 183.21.1.0/24.
- Configure IPv4 and IPv6 default route in R3 using the exit interface command.

- Configure IPv4 and IPv6 default route in R4 using the next-hop command.

Task 5 Application Server

Configure secure web services on web server. Then, configure the email server to send and receive email. Add TWO users (**U2020**, **U2021**) with password cisco in the email server. Set the domain name to **um.edu.my**.

Ensure that all PCs can ping to the servers.

Once complete the configuration, upload the **pka file** using the link given in the question paper. Enter your name, matric number (**17171717 / 25001717 / S2101717**), and mark. To avoid grading penalties, please double-check your submission. **The examiner is not responsible for files that are blank, corrupted, or incorrect.**

<https://docs.google.com/forms/d/e/1FAIpQLSf9iaz2Wp1CBcpJXPXXhhQrr5YWZ0RHUtbuZeNPB2szBdoSgA/viewform?usp=dialog>

If you encounter technical issues with the link provided, please attempt your submission using an **alternative email account**. Please note: **Submissions sent via email will not be accepted or graded.**

Please ensure all work is submitted **on time**. Late submissions will incur a marks penalty, and the link will be permanently disabled **10 minutes after the deadline**.

Please do not copy answer to your course mates. The AI grading system will be able to detect and will result in **a score of zero for all parties involved**.